

Department of Electrical and Electronics Engineering

University of Dhaka

Batch - 103, EEE-58

2nd Year 1st Semester



Course: EEE-2101

Title: Analog Electronics I

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SYLLABUS

PN Junction Diode and Diode Circuits: Semiconductor Materials, Construction of PN Junction, Formation of Depletion Layer and Barrier Voltage in PN Junction, Current-Voltage Characteristics of a PN Junction Diode, Equivalent Circuit of Diode, Resistance and Capacitance of PN Junction, Diode Circuit and Load Line, Zener Diode Operation and Application, Half-Wave and Full-Wave Rectifier, Voltage Multiplier, Clipper and Clamper.

Bipolar Junction Transistors (BJT): Construction and Operation of BJT, Amplifying Action, Characteristics of BJT in Common Base (CB), Common Emitter (CE) and Common Collector (CC) Configurations, α and β , Q-Point and Load Line, Different Biasing Circuits, Stability Factor, BJT as a Switch, re and Hybrid Equivalent Circuit of BJT.

Single Stage BJT Amplifier Circuits: Operation of Single-Stage Amplifier, Voltage and Current Gain, Input and Output Impedance of CB, CE, CC Configurations using h-Parameter, Effect of Unbypassed R_E , R_S .

Field Effect Transistor (FET): JFET Structure, Operation and Characteristics, hParameters for JFET, MOSFET Construction, Operation and Characteristics, Biasing Circuits of JFET and MOSFET, Single-stage JFET Amplifier, MOSFET as a Switch, CMOS Inverter.

Multistage and Differential Amplifiers: RC Coupled Two Stage Amplifiers, Direct-coupled Amplifier, Darlington Pair, Multistage Frequency Effects.

Feedback Amplifiers: Principle of Feedback Amplifier, Positive and Negative Feedback, Advantages of Negative Feedback, Gain Stability, Decreased Distortion, Increased Bandwidth, Forms of Negative Feedback, Practical Feedback Circuits.

REFERENCE BOOKS

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4. Electronic Principles, Paperback Edition, Albert Malvino and David Bates, McGraw Hill.
5. Electronic Circuits: Discrete and Integrated, Paperback Edition, Donald Schilling, Charles Belove, and Raymond Saccardi, McGraw Hill.
6. Microelectronics, 2nd Edition, Jacob Millman and Arvin Grabel, McGraw Hill.
7. Microelectronic Circuits: Theory and Applications, 7th Edition, Adel S. Sedra, Kenneth C. Smith and Arun N. Chandorkar, Oxford University Press.

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CHAPTER 1

PN JUNCTION DIODE AND DIODE CIRCUITS

1.1 Semiconductor Materials

Conductor: The term conductor is used for any material that will support generous flow of charge when the voltage is applied across its terminals.

Insulator: An insulator is a material that offers very low conductivity when voltage is applied.

Semiconductor: It is a material that has conductivity more than insulators but less than the conductor. Ex: Si, Ge

Resistivity and Conductivity

$$\rho = \frac{1}{\sigma} = \frac{RA}{L}$$

Where,

ρ = Resistivity

σ = Conductivity

A = Area of the cross-section

R = Resistance

L = Length

According to Ohm's Law: $R \uparrow I \downarrow$ and $R \downarrow I \uparrow$

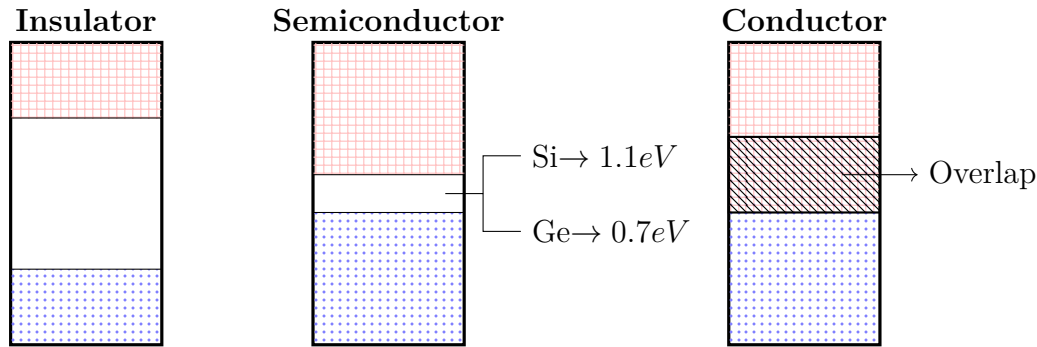
So,

$\rho_c < \rho_s < \rho_i$ $\rho \longrightarrow$ Resistivity

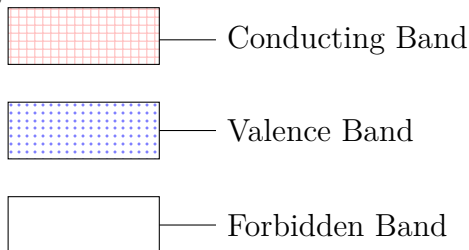
$\sigma_c > \sigma_s > \sigma_i$ $\sigma \longrightarrow$ Conductivity

$I_c > I_s > I_i$ $I \longrightarrow$ Current

Energy Band Diagram



Here,



Crystal Structure of Silicon

- Silicon crystallize in a diamond cubic lattice. Each Si has 4 valence electrons. To achieve stability, each atom shares 1 valence electron with each of its four neighbours.

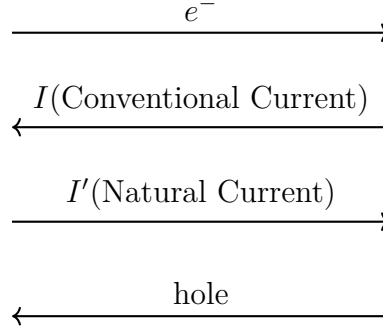
Intrinsic Semiconductor

- They are the pure semiconductors. No other atoms included.
- Free electrons are only due to natural causes.

Extrinsic Semiconductor

- Impurity atoms are added.
- Two types of impurity
 - Pentavalent Impurity: Added atoms like Antimony, P, As
 - Trivalent Impurity: Added atoms like B, Al
- Added 1 part in 10 millions.
- **Doping:** Process of adding certain impurity atoms to the pure semiconductors is called doping.
- There are **2** types of extrinsic semiconductor.
 - **n-type:** we added pentavalent impurity.
 - **p-type;** we added trivalent impurity.

Directions:



Properties Name	Si(Silicon)	Ge (Germanium)
Energy Gap (eV)	1.1	0.7
Electron Mobility ($m^2/V.s$)	0.135	0.39
Hole Mobility ($m^2/V.s$)	0.048	0.19
Intrinsic Carrier Density (n_i)	1.5×10^{16}	2.4×10^{19}
Intrinsic Resistivity	2300	0.46
Density (gm/m^3)	2.33×10^6	3.32×10^6

1.2 Mass Action Law

Statement: Under thermal equilibrium, the product of the free electrons concentration and the free hole concentration is equal to a constant to the square of intrinsic carrier concentration.

$$n.p = n_i^2 \quad \left| \begin{array}{l} n = \text{Free electron concentration} \\ p = \text{Free hole concentration} \\ n_i = \text{Intrinsic Carrier Concentration} \end{array} \right.$$

In Intrinsic Semiconductor,

$$\text{no of } e^- = \text{no of holes}$$

$$n = p = n_i$$

1.3 Charge Density in a Semiconductor

- Electrical neutrality is satisfied by semiconductor also.

$$\text{total } \oplus \text{ charge} = \text{total } \ominus \text{ charge}$$

$N_D + p = N_A + n$	$N_A =$ Concentration of Acceptor ion
	$N_D =$ Concentration of Donor ion
	$p =$ Hole Concentration
	$n = e^-$ Concentration

Case 1: Consider n-type material

Here,	$n \gg p$ $N_D = n$ $n_n \simeq N_D$
$N_A = 0$	
So,	
$N_D + p = n$	
$\Rightarrow N_D = n - p$	

$$n \cdot p = n_i^2$$

$$\Rightarrow n_n \cdot p_n = n_i^2$$

$$\Rightarrow N_D \cdot p_n = n_i^2$$

$$\Rightarrow p_n = \frac{n_i^2}{N_D}$$

Case 2: Consider p-type material

Here,	$p \gg n$ $p = N_A$ $p_p \simeq N_A$
$N_D = 0$	
So,	
$N_A + n = p$	
$\Rightarrow N_A = p - n$	

$$n \cdot p = n_i^2$$

$$\Rightarrow n_p \cdot p_p = n_i^2$$

$$\Rightarrow n_p \cdot N_A = n_i^2$$

$$\Rightarrow n_p = \frac{n_i^2}{N_A}$$

1.4 Conductivity of Semiconductor

- Metals are unipolar. But semiconductors are bipolar.
- When Electric Field E is applied, e^- and holes are drifted in opposite direction but the current due to them will be in the same direction.

$v_d \propto E$	$\mu = \text{Mobility}$
$\Rightarrow v_d = \mu E$	$i = \text{Current}$
$i = neAv_d$	$n = \text{no. of charge carrier}$
$\sigma = ne\mu$	$A = \text{Area of cross-section}$
$= (n\mu_n + p\mu_p)e$	$J = \text{Current Density}$
$J = \frac{i}{A}$	$\sigma = \text{Conductivity}$
$= nev_d$	$E = \text{Electric Field}$
$= ne\mu E$	$v_d = \text{Drift Velocity}$
$= \sigma E$	$e = \text{Charge of a } e^-$
$= (n\mu_n + p\mu_p)eE$	$e.m = \text{Effective Mass}$
	$\mu \propto \frac{1}{e.m}$
	$e.m_{hole} \gg e.m_{e^-} \quad \mu_p \ll \mu_n$

Example 1.1: The mobilities of free electron and holes in a pure germanium are 0.38 and $0.18 \text{ m}^2/\text{V.s}$. Find the value of intrinsic conductivity. Assume $n_i = 2.5 \times 10^{19}/\text{m}^3$ at room temperature.

Given,

Mobility of free electron, $\mu_n = 0.38/\text{m}^3$

Mobility of holes, $\mu_p = 0.18/\text{m}^3$

$\therefore$ This is a pure germanium semiconductor.

$$\therefore n = p = n_i = 2.5 \times 10^{19}/\text{m}^3$$

We know that,

$$\begin{aligned} \text{Conductivity, } \sigma &= (n\mu_n + p\mu_p)e \\ &= (0.38 + 0.18) \times 2.5 \times 10^{19} \times 1.6 \times 10^{-19} (\Omega - m)^{-1} \\ &= 2.24 (\Omega - m)^{-1} \text{ (ans)} \end{aligned}$$

Question 1.1: In a germanium sample, a donor type impurity is added to the extent of 1 atom per 10^8 germanium atoms. Show that resistivity of the germanium sample drops to 3.7 ohm-cm

Given,

$$\mu_n = 3800 \text{ cm}^2/\text{V.s} \quad n_i = 2.5 \times 10^{13}/\text{cm}^3$$

$$\mu_p = 1800 \text{ cm}^2/\text{V.s} \quad N_{ge} = 4.41 \times 10^{22}/\text{cm}^3$$

Question 1.2: Consider an intrinsic silicon bar of cross-section 5cm^2 and length 0.5cm at room temperature 300K . An average field of $20\text{V}/\text{cm}$ is applied across the ends of the silicon bar.

Calculate:

- Electron and hole components of current density.
- Total current in the bar.
- Resistivity of the bar.

Given,

$$\mu_n = 1400\text{cm}^2/\text{V.s} \quad n_i = 1.5 \times 10^{10}/\text{cm}^3$$

$$\mu_p = 450\text{cm}^2/\text{V.s}$$

1.5 PN Junction Diode (No Applied Bias)

- Bias means application of external voltage across the two terminals.
- BJT means Bipolar Junction Transistor (3 terminals device, having 2 diodes)
- PN Junction constitutes a rectifier which permits the easy flow of charge in one direction but restrains the flow in the opposite direction.
- p-type
 - Majority Charge Carrier: hole
 - Minority Charge Carrier: e^-
- n-type
 - Majority Charge Carrier: e^-
 - Minority Charge Carrier: hole
- Diffusion current $\longrightarrow$ because of Majority Charge Carrier
- Immobile ions will surface out because of diffusion, it it depletion region.
- Depletion Region/Space Charge Region: No mobile charge but only uncovered immobile ions.
- Width of depletion region is **fixed**.
- Drift current $\longrightarrow$ because of Minority Charge Carrier
- Diffusion current have the opposite direction of drift current.
- Under Steady State:

$$\text{Diffusion Current} = \text{Drift Current}$$

$$\text{Net current} = 0$$

1.6 Barrier Potential

Barrier Potential/ Built-in Potential,

$$\begin{aligned} V_b &= \frac{kT}{e} \ln \left(\frac{N_A N_D}{n_i^2} \right) \\ &= V_T \ln \left(\frac{N_A N_D}{n_i^2} \right) \end{aligned}$$

Here,

k = Boltzman constant = $1.38066 \times 10^{-23} J/K$

T = Absolute Temperature

e = Charge of one e^- = $1.6 \times 10^{-19} C$

At room temperature (300K/27°C):

$$\begin{aligned} V_T &= \frac{kT}{e} \\ &= \frac{1.38066 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} \\ &= 0.026V \end{aligned}$$

- In ideal cases, we consider the barrier potential for Si is 0.7V and 0.3V is for Ge at room temperature.

Example 1.2: Consider a pn junction at room temperature, doped as $N_A = 10^{16}/cm^3$ in the p-region and $N_D = 10^{17}/cm^3$ in the n-region. Intrinsic carrier density is $1.5/cm^3$ at room temperature. Calculate the built-in potential.

Given,

Acceptor ions Concentration, $N_A = 10^{16}/cm^3$

Donor ions Concentration, $N_D = 10^{17}/cm^3$

At room temperature, $V_T = 0.026V$

Intrinsic Charge density, $n_i = 1.5/cm^3$

We know that,

$$\begin{aligned} \text{Built-in Potential, } V_b &= V_T \ln \left(\frac{N_A N_D}{n_i^2} \right) \\ &= 0.026 \ln \left\{ \frac{10^{16} \times 10^{17}}{(1.5 \times 10^{10})^2} \right\} \\ &= 0.757V \end{aligned}$$

1.7 Width of Depletion Layer

Width of Depletion Region,

$$W_d = \sqrt{\frac{2\varepsilon}{e} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_b}$$

$$\varepsilon = \varepsilon_r \varepsilon_0 \quad \left| \begin{array}{l} \varepsilon = \text{Electric Permittivity} \\ \varepsilon_r = \text{Relative Permittivity} \\ \varepsilon_0 = \text{Permittivity of Vacuum} \end{array} \right.$$

- For Si
 - $\varepsilon = 1.04 \times 10^{-12} \text{ F/cm}$
 - $0.1 \mu\text{m} < W_d < 1 \mu\text{m}$
- For Ge, $\varepsilon = 1.12448 \times 10^{-12} \text{ F/cm}$

Example 1.3: Consider a silicon pn-junction at room temperature doped at $N_A = 10^{16}/\text{cm}^3$ in the p-region and $N_D = 10^{17}/\text{cm}^3$ in the n-region. Intrinsic carrier density is $1.5/\text{cm}^3$ at room temperature. Calculate the width of depletion region.

Given,

Acceptor ions Concentration, $N_A = 10^{16}/\text{cm}^3$

Donor ions Concentration, $N_D = 10^{17}/\text{cm}^3$

At room temperature, $V_T = 0.026 \text{ V}$

Intrinsic Charge density, $n_i = 1.5/\text{cm}^3$

Permittivity of Si, $\varepsilon = 1.04 \times 10^{-12} \text{ F/cm}$

We know that,

$$\begin{aligned} \text{Built-in Potential, } V_b &= V_T \ln \left(\frac{N_A N_D}{n_i^2} \right) \\ &= 0.026 \ln \left\{ \frac{10^{16} \times 10^{17}}{(1.5 \times 10^{10})^2} \right\} \\ &= 0.757 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Width of depletion region, } W_d &= \sqrt{\frac{2\varepsilon}{e} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) \cdot V_b} \\ &= \sqrt{\frac{2 \times 1.04 \times 10^{-12}}{1.6 \times 10^{-19}} \left(\frac{1}{10^{16}} + \frac{1}{10^{17}} \right) \times 0.757} \\ &= 3.290 \times 10^{-5} \text{ cm} \\ &= 0.3290 \mu\text{m} \end{aligned}$$

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